

$$\gamma \sim N(0, V_\gamma/p)$$

$$y \mid \gamma \sim N(\mathbf{H}\gamma, V_e)$$

Define

$$V_\ell := \text{Var}(\mathbf{H}\gamma) \neq V_\gamma$$

if you have LD scores and $\hat{V}_\gamma$ can you approximate $\hat{V}_\ell \leftarrow$ we want this.

If not, might still be okay. In partitioned model, could compare $V_{\gamma,1}$ to $V_{\gamma,2}$ or something for different annotations.

Decompose $V_\ell := \text{Var}(\mathbf{H}\gamma)$

Let $\mathbf{Z}$ be the column-standardized genotype matrix and $\mathbf{H} \in \mathbb{R}^{n \times p}$, $p = \binom{m}{2}$, the unstandardized epistatic design matrix with k -th column $\mathbf{H}_{.k} = \mathbf{Z}_{.i} \circ \mathbf{Z}_{.j}$ for the pair (i, j) , $i < j$. Effects are drawn as $\gamma \sim N(0, p^{-1}V_\gamma \mathbf{I}_p)$, and V_ℓ is the realized variance of $g = \mathbf{H}\gamma$ across individuals.

Writing $\mathbf{P} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^T$ for the centring projection, expend the variance

$$\begin{aligned} V_\ell &= \text{Var}(\mathbf{H}\gamma) \\ &= \frac{1}{n} \gamma^T \mathbf{H}^T \mathbf{P} \mathbf{H} \gamma \end{aligned}$$

Take expectation of γ : $\mathbb{E}_\gamma[\gamma^T A \gamma] = \mathbb{E}_\gamma[\text{tr}(\gamma^T A \gamma)] = \mathbb{E}_\gamma[\text{tr}(A \gamma \gamma^T)] = \text{tr}(A \mathbb{E}(\gamma \gamma^T)) = \frac{V_\gamma}{p} \text{tr}(A)$

$$\begin{aligned} &= \frac{V_\gamma}{np} \text{tr}(\mathbf{H}^T \mathbf{P} \mathbf{H}) \\ &= \frac{V_\gamma}{np} \sum_{k=1}^p [\mathbf{H}^T \mathbf{P} \mathbf{H}]_{kk} \\ &= \frac{V_\gamma}{np} \sum_{k=1}^p n \widehat{\text{Var}}(\mathbf{H}_{.k}) \\ &= \frac{V_\gamma}{p} \sum_{k=1}^p \widehat{\text{Var}}(\mathbf{H}_{.k}) \\ &= \frac{V_\gamma}{p} \sum_{i < j} \widehat{\text{Var}}(Z_i \circ Z_j) \end{aligned}$$

Therefore, $V_\ell = V_\gamma \left(\frac{1}{p} \sum_{i < j} \widehat{\text{Var}}(Z_i \circ Z_j) \right) = c V_\gamma$.

Now we are thinking about can we represent $c = \frac{1}{p} \sum_{i < j} \widehat{\text{Var}}(Z_i Z_j)$ using LD score.

Closed form for c

Under HWE the per-pair variance has a closed form in the allele frequencies and the pairwise correlation (error order $\left(\frac{1}{\sqrt{n}}\right)$ as using sample variance). site from previous notes

$$\begin{aligned} \text{Var}(Z_i Z_j) &= 1 + \frac{\sigma_{ij}(4p_i p_j - 2p_i - 2p_j + 1)}{4(p_i p_j - p_i p_j^2 - p_j p_i^2 + p_i^2 p_j^2)} \\ &= 1 + \frac{\sigma_{ij}(1 - 2p_i)(1 - 2p_j)}{4p_i p_j (1 - p_j)(1 - p_i)} \end{aligned}$$

replace σ_{ij} with r_{ij} :

$$\text{Var}(Z_i Z_j) = 1 + r_{ij} s_i s_j, \quad s_i = \frac{1 - 2p_i}{\sqrt{2p_i(1 - p_i)}},$$

where s_i is the skewness of the standardized genotype at SNP i . Substituting,

$$\begin{aligned} c &= \frac{1}{p} \sum_{i < j} \text{Var}(Z_i Z_j) \\ &= \frac{1}{p} \sum_{i < j} (1 + r_{ij} s_i s_j) \end{aligned}$$

the def of LD score: $l_i = \sum_k r_{ik}^2$, we may not able to insert LD score into the c . but can use this estimation to do:

$$\hat{c} = \frac{1}{p} \sum_{i < j} (1 + \hat{r}_{ij} \hat{s}_i \hat{s}_j)$$

Written pairwise, $\hat{c}$ has $\binom{m}{2}$ terms and costs $O(nm^2)$ time with $O(m^2)$ storage for $\mathbf{R}$. Neither is necessary: since $\mathbf{R}$ is symmetric with unit diagonal,

$$\sum_{i < j} \hat{r}_{ij} \hat{s}_i \hat{s}_j = \frac{1}{2} (\hat{s}^T \mathbf{R} \hat{s} - \|\hat{s}\|^2), \quad \hat{s}^T \mathbf{R} \hat{s} = \frac{\|\mathbf{Z} \hat{s}\|^2}{n},$$

so $\hat{c}$ follows from a single matrix–vector product in $O(nm)$ time and $O(n)$ storage. Meanwhile, calculate s takes $O(nm)$ as well.

The logic

We use raw H in REML, which will give the estimation $\hat{V}_\gamma$, we want $V_\ell := \text{Var}(\mathbf{H}\gamma)$. We find that

$$\mathbb{E}[V_\ell] = V_\gamma * c \quad \text{where } c = \frac{1}{p} \sum_{i < j} \text{Var}(Z_i Z_j)$$

In simulation, the first noise come from the sampling error of realized variance V_ℓ . Compute c using genotype data cost $O(m^2 n)$ time, so we first calculate AF and use HWE to compute c in $O(mn)$ time.

Two noise come from sampling error and HWE rule.